

Chapter 4: Functions, Relations & Graphing

Domain/Range, Function Composition, Inverses, Transformations & Transformations

4.1 Concept of a Function & Domain Analysis

A function $f: X \rightarrow Y$ assigns exactly one output $f(x) \in Y$ to each input $x \in X$. The set of valid inputs is the **Domain**; the set of outputs is the **Range**.

Domain Restrictions

1. **Denominators:** Cannot equal zero.
2. **Even Roots:** Radicand must be ≥ 0 .
3. **Logarithms:** Argument must be > 0 .

Function Classifications

- **Injective (1-to-1):** Passes horizontal line test.
- **Surjective (Onto):** Range equals codomain.
- **Bijjective:** Both 1-to-1 and onto (Invertible).

Worked Example 4.1: Finding Domain

Problem: Find the domain of $f(x) = \sqrt{x-3} / (x-7)$.

Solution:

1. Radicand requirement: $x - 3 \geq 0 \Rightarrow x \geq 3$.
 2. Denominator requirement: $x - 7 \neq 0 \Rightarrow x \neq 7$.
- Combining conditions: $[3, 7) \cup (7, \infty)$.

4.2 Function Composition & Inverse Functions

Composite function: $(f \circ g)(x) = f(g(x))$. An inverse function $f^{-1}(x)$ satisfies $f(f^{-1}(x)) = x$.

Steps to find $f^{-1}(x)$:

1. Replace $f(x)$ with y : $y = f(x)$
2. Swap x and y : $x = f(y)$
3. Solve explicitly for y , then write $y = f^{-1}(x)$.

4.3 Graphical Transformations

Transforming a base graph $y = f(x)$ allows rapid sketching of complex functions without pointwise plotting.

Rigid Transformations

- *Vertical Shift:* $y = f(x) + c$ (up c units)
- *Horizontal Shift:* $y = f(x - c)$ (right c units)
- *Reflection over x -axis:* $y = -f(x)$
- *Reflection over y -axis:* $y = f(-x)$

Non-Rigid Transformations

- *Vertical Stretch:* $y = a \cdot f(x)$ ($a > 1$)
- *Vertical Compression:* $y = a \cdot f(x)$ ($0 < a < 1$)
- *Horizontal Compression:* $y = f(c \cdot x)$ ($c > 1$)
- *Horizontal Stretch:* $y = f(x/c)$ ($c > 1$)

Worked Example 4.2: Finding Inverses

Problem: Find the inverse function of $f(x) = (2x + 1) / (x - 3)$.

Solution: Set $y = (2x + 1)/(x - 3)$. Swap x and y :

$$x = (2y + 1) / (y - 3) \Rightarrow x(y - 3) = 2y + 1 \Rightarrow xy - 3x = 2y + 1$$

$$\text{Group } y \text{ terms: } xy - 2y = 3x + 1 \Rightarrow y(x - 2) = 3x + 1$$

$$f^{-1}(x) = (3x + 1) / (x - 2).$$

Chapter 4 Review & Practice Problems

1. Find the domain of $g(x) = \sqrt{16 - x^2}$.
2. Given $f(x) = x^2 + 3$ and $g(x) = \sqrt{x - 1}$, find $(f \circ g)(x)$ and $(g \circ f)(x)$.
3. Determine whether $f(x) = 3x - 7$ is bijective, and if so, find $f^{-1}(x)$.
4. Describe the sequence of transformations required to obtain the graph of $y = -2(x - 4)^2 + 5$ from $y = x^2$.

